MIL-H-43208A (GL)

4 November 1969

SUPERSEDING
MIL-H-43208 (GL)
17 March 1964

MILITARY SPECIFICATION

HANGER, PAPER HOLDING

1. SCOPE

1.1 This specification covers one type and size paper-holding hanger for conditioning printing paper.

* 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-C-843	-	Cushioning Material, Cellulosic.

MILITARY

MIL-B-117	-	Bags, Sleeves and Tubing, - Interior Packaging.

STANDARDS

FEDERAL

FED-STD-595	-	Colors.

MILITARY

MIL-STD-105	-	Sampling Procedures, and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-130	-	Identification Marking of U.S. Military Property.

FSC 3610

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DRAWING

U.S. ARMY NATICK LABORATORIES

6-1-5505 - Hanger, Paper Holding.

(Miniature copy of Drawing 6-1-5505, identified as figure 1, is attached for information only.)

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following publications of the issues in effect on date of invitation for bids or request for approval form a part of this specification to the extent specified herein:

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to American Trucking Associations, Attn: Tariff Order Section, 1616 P Street, N.W., Washington, D.C. 20036.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 202 Union Station, 516 W. Jackson Blvd., Chicago, Illinois 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.)

* 3. REQUIREMENTS

3.1 First article.- When specified (see 6.2), before production is commenced, a preproduction sample paper holding hanger (rack), conforming to this specification shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. Approval of the preproduction sample authorizes the commencement of production but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

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3.2 Materials.- Materials and components not definitely specified shall be of the quality normally used by the manufacturer for paper hangers, provided the end item complies with all provisions of this specification.

3.3 Design.- The rack shall be designed for overhead or floor mounting, without change of design or appurtenances, and shall have a capacity to hold from one to 3,000 sheets of 35 inch by 45 inch substance 60 (500 - 25 by 38 inch sheets) book paper, equally divided between 50 to 60 pairs of paper hangers (see 4.4). The end item shall be assembled by means of threaded joints and fittings, and shall permit disassembly, and subsequent reassembly, without the need for special tools or modification. Anchoring of the assembly shall be by means of floor flanges threaded onto not less than 6 vertical supports. The mounted assembly shall be capable of withstanding not less than 5 seconds of manual shaking without visible evidence of deformation, structural failure, or loss of paper from the hangers, when tested in accordance with 4.4.

3.4 Construction.- The rack shall be fabricated from ferrous or nonferrous metal, and shall conform to the requirements shown on Drawing 6-1-5505, and as specified herein. Additional bracing will be permitted wherever considered necessary to meet performance requirements specified herein, providing such additions do not obstruct accessibility to the paper hangers, nor decrease the paper holding capacity of the rack. All like components shall be interchangeable.

3.4.1 Details of components.- The component parts shown on Drawing 6-1-5505 shall conform to the following additional requirements.

3.4.1.1 Vertical supports and cross connectors.- Vertical supports and cross connectors shall be fabricated from minimum 1-1/4 inch diameter pipe. The length of the cross connectors shall permit adjustment of the hanger strips from a minimum of 9 inches to a maximum of 35 inches, center to center. The intermediate cross connectors shall be located midway between the floor flanges and the cross connectors that carry the hanger strips.

3.4.1.2 Floor flanges.- The floor flanges shall be commercially available threaded flanges provided with mounting holes.

3.4.1.3 Hanger strips.- The hanger strips, which support the paper hangers, shall be fabricated of commercially available T-bar. The metal thickness shall be minimum 1/8 inch. The flange shall be minimum 1-1/4 inches. Minimum 1/4 inch diameter holes shall be provided for attachment of the paper hangers.

3.4.1.4 Paper hangers.- Not less than 50 nor more than 60 paper hangers shall be provided on each of the two hanger strips. The paper contact surfaces shall be corrosion-resisting steel, copper alloy, or cast malleable iron plated to resist corrosion. The paper hangers shall be bolted to the hanger strips so that opposing hangers are parallel to each other in the assembled rack, for use in pairs. The hanger tongues shall be secured to the hangers in such a manner to afford free movement of the tongue without locking, binding, marking of paper, or requiring manipulation of the tongues to load 25 or more sheets of paper (see 4.4).

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3.4.1.5 Locking devices.- The locking devices shall be an integral part of the hanger assembly, and shall be equipped with 1/4 inch square-headed set screws for use in securing the hanger strips the desired distance from center to center (see 3.4.1.1 and 4.4).

3.5 Finish.-

3.5.1 Pretreatment.- Prior to enameling, all component parts, shall be clean and free from all foreign matter, rough edges, and surfaces.

3.5.2 Enameling.- Olive drab enamel conforming to color No. X24087 of FED-STD-595, shall be applied to all exterior surfaces of component parts, except where function may be adversely affected. Where the enamel coat is omitted, the surfaces shall be corrosion-resisting steel, copper alloy, or cast malleable iron plated to resist corrosion.

3.6 Identification marking.- The rack shall be marked with the letters "U.S.", and the manufacturer's name or readily identifiable trademark, in accordance with MIL-STD-130. Marking shall be located as shown on Drawing 6-1-5505.

3.7 Assembled instructions.- Comprehensive assembly instructions shall be furnished with each rack. The instructions shall be an exploded view type of drawing, typed or printed instructions, or a combination of both.

3.8 Workmanship.- The end item shall be clean and free of corrosion, snags, and rough and sharp edges. All functioning components of the assembled rack shall operate smoothly without binding. There shall be no loose (see 4.4), missing, or misplaced components. Painted and plated surfaces shall be uniformly coated and dry and free from runs, sags, teardrops, foreign matter, chips, scratches, blisters, flaking, peeling, and areas of no film.

* 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 First article inspection.- When a preproduction sample is required, it shall be examined for defects in tables I and II, dimensions specified, and tested as specified in 4.4. The presence of any visual defect, any dimension not within the specified requirements, or failure to pass the test shall be cause for rejection of the preproduction sample.

4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection.- In accordance with 4.1 above, the supplier is responsible for insuring that components and materials used are manufactured, tested, and inspected in accordance with all the requirements of referenced specifications, drawings, and standards, unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.3.2 In-process inspection.- Examination shall be made during the manufacturing process to determine compliance with the enameling pretreatment requirements of 3.5.1, and the requirements for paper contact surfaces in 3.4.1.4. Whenever a deviation is noted, correction shall be made to affected items and process.

4.3.3 End item inspection.- The lot shall be all knocked-down racks offered for inspection at one time. The sample unit shall be all of the component parts required to erect one rack.

4.3.3.1 Visual examination.- The knocked-down racks shall be examined for defects in table I. The inspection level shall be II with an acceptable quality level (AQL) of 4.0 expressed in terms of defects per hundred units.

TABLE I.- Visual examination of knocked-down racks

Examine	Defect
Finish	Rust or indication of rust under paint
	Nottype or color specified
	Poor adhesion, i.e., peeling, blistered, or flaking
	Tacky, runs, or sags
	Area of no film
	Not smooth
	Discoloration, stain, or dirt
	Color separation
	Foreign matter
	Color shade varies between components

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TABLE I.- Visual examination of knocked-down racks (cont'd)

Examine	Defect
Construction and workmanship	Any component part not as specified Any break, fracture, or crack Sharp edges, slivers, or burrs Any component malformed, buckled, warped, out of shape, or damaged Like components not interchangeable
Assembly instructions	Missing or illegible
Marking for identification	Missing, illegible, incorrect, or in wrong location

* 4.3.3.2 Assembly examination.- One knocked-down rack shall be assembled in accordance with the assembly instructions and examined for defects in table II. The presence of any defect shall be cause for rejection of the lot.

TABLE II.- Assembly defects

Examine	Defect
Assembly	Requires special tools to assemble. Requires modification to assemble. Any component or part missing. Assembly out of square. Requires excessive force to assemble. Any component or part missing. Paper hanger tongue binds or is poorly secured. Opposing hanger not in line. Center distance between hanger strips not adjustable. Hanger strip binds on cross connector, or cannot be locked in desired location.
Assembly instructions	Incorrect.

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* 4.3.3.3 Dimensional examination.- The assembled rack shall be examined to determine compliance with overall dimensions specified on Drawing 6-1-5505, and the center to center adjustment range of hanger strips (see 3.4.1.1). Any nonconformance shall be cause for rejection of the lot.

* 4.3.3.4 Testing of the end product.- When a preproduction sample is not required, the initial unit of production shall be tested as specified in 4.4. Failure to pass the test shall be cause for rejection of the lot.

4.3.4 Examination of preparation for delivery.- An examination shall be made to determine that packaging, packing, and marking as required by section 5 of this specification are complied with. Defects shall be as indicated in table III. The sample unit shall be one shipping container fully prepared for delivery. The lot shall be all of the containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0, expressed in terms of defects per hundred units.

TABLE III.- Examination of preparation for delivery

Examination	Defect
Markings	Omitted; incorrect; illegible; of improper size, location, sequence or method of application.
Materials	Component missing or damaged or otherwise defective.
Workmanship	Inadequate application of components, such as cushioning and blocking, and bracing, inadequate or improperly affixed.

4.3.4.1 Examination of shipping containers.- Examination for defects in closure and strapping of shipping containers shall be in accordance with the appendix of the applicable container specification.

4.4 Test methods.- The rack shall be assembled and secured to the floor. The rack shall then be tested by loading each pair of hangers with 1, 25, 30, and 50 or 60 (3,000 total) sheets of 35 by 45 inch book paper conforming to 3.3. After each loading, the rack shall be manually shaken for 5 seconds. The test shall then be repeated with the rack mounted overhead. Evidence of any one of the following shall constitute failure of this test:

(a) Failure to receive the quantity of paper (see 3.3).
(b) Manipulation of hanger tongue required to load 25 or more sheets of paper per pair of hangers (see 3.4.1.4).

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(c) Paper marked by paper hanger (see 3.4.1.4).
(d) Release of any paper before, after, or during testing (see 3.3).
(e) Structural failure or distortion of the assembly (see 3.3).
(f) Loosening of any component or locking device (see 3.4.1.5 and 3.8).

5. PREPARATION FOR DELIVERY

5.1 Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.- The two hanger strips shall be removed from the paper-holding hanger. All pipe components of the rack shall be disassembled into straight lengths. The assembly instructions shall be placed in a sealed bag conforming to type III, class B, style optional of MIL-B-117.

5.1.2 Level C.- Paper-holding hangers shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing.- Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.- Each paper-holding hanger, packaged as specified in 5.1, shall be packed in the most compact manner in a nailed wood or cleated plywood shipping container conforming to class 2, style 2 or 4, type 3 load of PPP-B-621; or overseas type, type 3 load of PPP-B-601, respectively. Lumber blocking and bracing shall be used to immobilize the contents within the shipping container. The blocking and bracing shall depend on endgrain nailing alone, but supplemental supporting cleats shall be provided to reinforce and prevent splitting of the blocking and bracing. The hanger strips shall be blocked in such a manner as to prevent movement or vibration of the tongues while in transit. Cushioning material conforming to type II, class B of PPP-C-843 or other water-resistant cushioning material shall be used as necessary to protect the finish of contents. The packaged assembly instructions shall be positioned at the top of the container. Each shipping container shall be provided with skids and closed and strapped in accordance with the applicable container specification or appendix thereto.

5.2.2 Level B.- Each paper-holding hanger, packaged as specified in 5.1, shall be packed as specified in 5.2.1, except that shipping containers shall conform to class 1, style 2 or 4, type 3 load of PPP-B-621, or domestic type, style A or B of PPP-B-601, and cushioning material need not be water-resistant.

5.2.3 Level C.- Paper-holding hangers, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

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5.3 Marking.- In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129. Opaque bags for assembly instructions shall be marked "Assembly Instructions".

6. NOTES

6.1 Intended use.- The paper-holding hanger covered by this specification is intended for use as a component part of a topographic reproduction set (FSN 3610-288-0301).

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) First article.- When a preproduction sample is required (see 3.1).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:

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User activity:

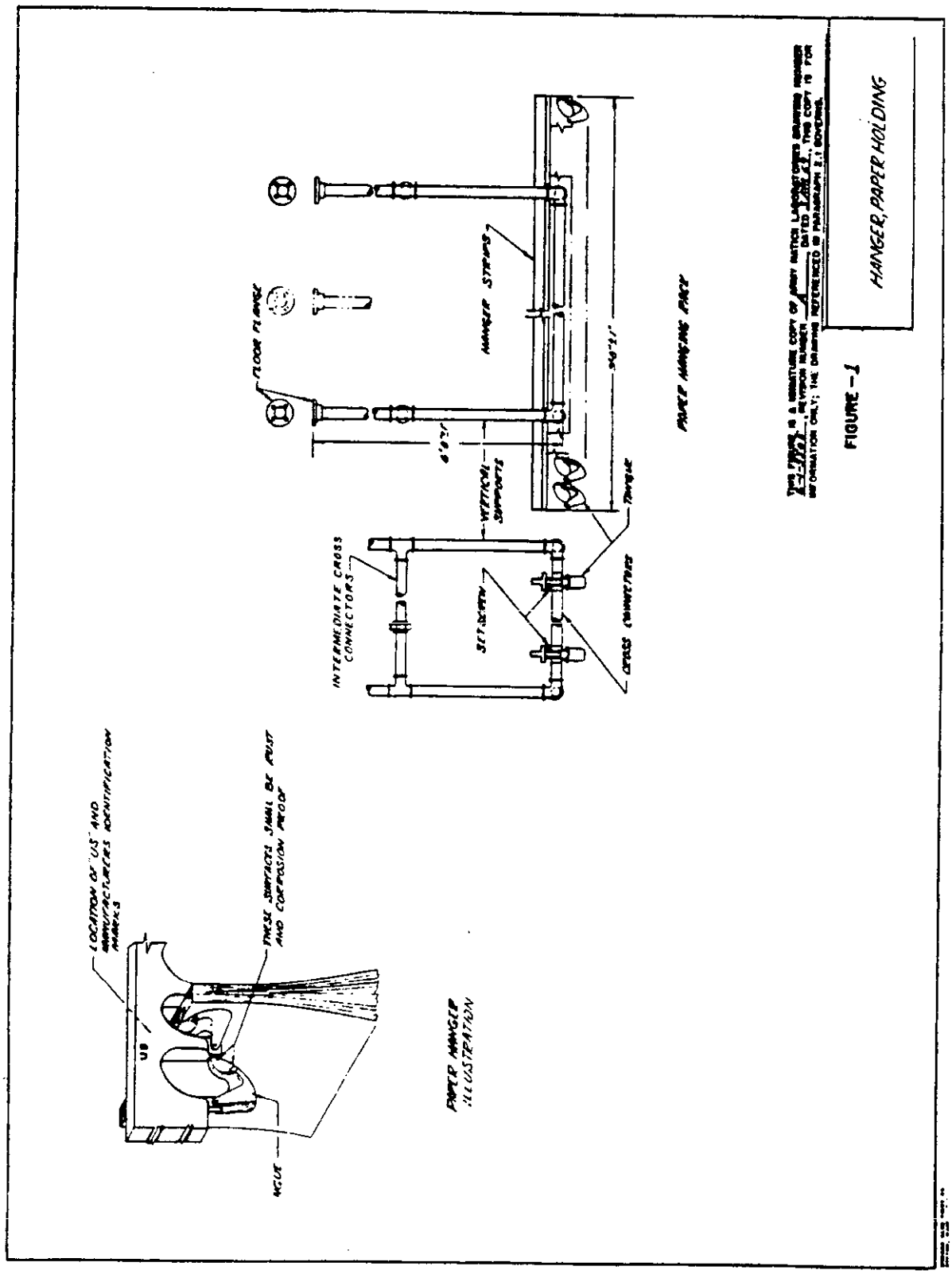
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Project No. 3610-A176

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
Hanger, Paper Holding MIL-H-43208A(GL)

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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